

MINERAL RESOURCES

Minerals are exhaustible, non-renewable resources found in the earth's crust. They are essential to our industrialized society and daily life. Due to rapid expansion of industries, the consumption of minerals has increased tremendously. A number of minerals such as silver, copper, mercury, tungsten etc. are now in short supply and likely to be exhausted within next 20 to 100 years.

Mining

The term mining refers to the process of taking out minerals or their ores from the earth.

Two Methods of mining: Surface mining and Sub-Surface mining

1. Surface Mining: This method of mining is utilized when mineral deposits occur at or near the surface of earth.

2. Sub-Surface Mining: This method is used when the mineral deposit lies deep beneath the earth's surface. Big holes are dug in the earth surface to extract out ores from the horizontal ore bodies.

Effects of Mining

1. The top soil is removed from the mining area to get access to the deposit. It disturbs and damages the land and results in spoiling the landscape.
2. Mining often causes ground subsidence, which results in tilting of buildings cracks in houses, buckling's of roads, bending of rail tracks and leaking of gas from cracked pipelines leading to serious disasters.
3. Mining disturbs the natural hydrological processes and also pollute groundwater as well as surface water. Sometimes radioactive substances like uranium also contaminate the water bodies through mine mine wastes.
4. Extraction and processing of ores emits enormous quantities of air pollutants such as suspended particulate matter leading to serious environmental hazards.
5. Miners often suffer from serious respiratory and skin diseases like asbestosis, silicosis, black lung disease etc due to constant exposure to the suspended particulate matter (SPM) and toxic substances.

Conservation of Minerals

1. **Recycling** – In recycling used and discarded items are collected, remelted and reprocessed into new products e.g. iron scraps, aluminium cans, etc. Some minerals present in products can be recycled e.g. gold, silver, lead, nickel, steel, copper, aluminium, zinc, etc.
2. **Reuse** – Certain items can be collected and can be used over and over again. e.g. reuse of glass bottles.
3. **Substitution** – The use of rare minerals can be substituted with more abundant minerals. For instance, in recent years, plastics, ceramics, high strength glass fibres and alloys have been substituted for rare materials like steel, tin and copper in many industries.
4. **Decreased Consumption** – Products that are durable and repairable should be used again instead of discarding them as waste.
5. **Use of waste** – The manufacturing industries may use the waste products of one manufacturing process as the raw materials for another industry.